Stochastic Processes — 1060429

Nawar Sgier, 326586047

Throughout, a Brownian motion (BM) is standard ($B_0 = 0$) with continuous paths, and $\mathcal{F}_t$ denotes the filtration in use. Apparent typos are pointed out where they occur.

Exercise 4

Stochastic calculus

Throughout, for $f \in L^2$ (adapted, with $\mathbb{E}\int_0^t f^2 ds < \infty$), $\int_0^t f\, dB$ is the L^2-limit of $\int f_n\, dB$ for simple adapted $f_n \to f$ in $L^2(ds \otimes d\mathbb{P})$. For a partition $0 = t_0 < \cdots < t_n = t$ of mesh $|\pi|$ write $\Delta_i = B_{t_i} - B_{t_{i-1}}$ and $\Delta t_i = t_i - t_{i-1}$.

1.

Remark. In (a)ii and (b)i the right-hand side should be $\frac{1}{3}B_t^3 - \int_0^t B_s\, ds$, not $\frac{1}{3}B_t^2 - \dots$. As stated it cannot hold: the left side has mean 0, while $\mathbb{E}[\frac{1}{3}B_t^2 - \int_0^t B_s\, ds] = \frac{t}{3}$. Itô's formula in (b) confirms the corrected form.

(a) i. $f_\pi = \sum t_{i-1}\mathbf{1}_{[t_{i-1},t_i)} \to s$ in L^2, since $\int_0^t |f_\pi(s) - s|^2 ds \le t|\pi|^2$. Summation by parts gives

$$\sum_i t_{i-1}\Delta_i = tB_t - \sum_i B_{t_i}\Delta t_i.$$

By continuity $|\sum_i B_{t_i}\Delta t_i - \int_0^t B_s\, ds| \to 0$ pathwise. It is bounded by $2t\sup_{[0,t]}|B| \in L^2$, so the convergence also holds in L^2. Hence $\int_0^t s\, dB_s = tB_t - \int_0^t B_s\, ds$.

(a) ii. For $u < s \le t$, by Cauchy–Schwarz and $B_s + B_u \sim N(0, s+3u)$, $\mathbb{E}(B_s^2 - B_u^2)^2 \le \sqrt{3(s-u)^2}\sqrt{3(4t)^2} = 12t(s-u)$. Hence $\sum_i B_{t_{i-1}}^2 \mathbf{1}_{[t_{i-1},t_i)} \to B^2$ in $L^2(ds \otimes d\mathbb{P})$, and $\int_0^t B^2 dB = \lim_{L^2} S_\pi$ with $S_\pi = \sum_i B_{t_{i-1}}^2 \Delta_i$. Summing $b^3 - a^3 = 3a^2(b-a) + 3a(b-a)^2 + (b-a)^3$ over the partition gives

$$S_\pi = \tfrac{1}{3}B_t^3 - \sum_i B_{t_{i-1}}\Delta_i^2 - \tfrac{1}{3}\sum_i \Delta_i^3.$$

The cross terms below vanish by independence of increments and since odd moments are 0, resp. $\mathbb{E}[\Delta_j^2 - \Delta t_j] = 0$:

$$\mathbb{E}\Big(\sum_i \Delta_i^3\Big)^2 = \sum_i 15\Delta t_i^3 \le 15t|\pi|^2, \qquad \mathbb{E}\Big(\sum_i B_{t_{i-1}}(\Delta_i^2 - \Delta t_i)\Big)^2 = \sum_i t_{i-1}\cdot 2\Delta t_i^2 \le 2t^2|\pi|.$$

Moreover $\sum_i B_{t_{i-1}}\Delta t_i \to \int_0^t B_s\, ds$ in L^2 as in i. Hence $\int_0^t B_s^2\, dB_s = \frac{1}{3}B_t^3 - \int_0^t B_s\, ds$.

(b) i. Itô's formula with $f(x) = x^3/3$ (so $f' = x^2$, $f'' = 2x$) gives $\frac{1}{3}B_t^3 = \int_0^t B_s^2\, dB_s + \int_0^t B_s\, ds$.

2.

Remark. We take $f, g \in L^2([0,T]\times\Omega)$, i.e. $\mathbb{E}\int_0^T f^2 < \infty$ for every T, the class on which the Itô integral is an L^2-martingale. If only $\int_0^T f^2 ds < \infty$ a.s. were assumed, the claim would be false in general. By Dudley's representation theorem (not part of the course and not used here) there is such an f with $\int_0^T f\, dB = 1$ a.s., and then $0 + \int_0^T f\, dB = 1 + \int_0^T 0\, dB$ with $C \ne D$.

The Itô integral has mean zero, so taking expectations gives $C = D$. Then $\int_0^T (f-g)\, dB = 0$ a.s., and by the Itô isometry

$$\mathbb{E}\int_0^T (f-g)^2 ds = \mathbb{E}\Big(\int_0^T (f-g)\, dB\Big)^2 = 0,$$

i.e. $f = g$ for $ds \otimes d\mathbb{P}$-a.e. (s, ω).

3.

(a) $d(B_t^2) = dt + 2B_t\, dB_t$.

(b) $d(2 + t + e^{B_t}) = (1 + \frac{1}{2}e^{B_t})\, dt + e^{B_t}\, dB_t$.

(c) $d(B_t^2 + \tilde{B}_t^2) = 2\, dt + 2B_t\, dB_t + 2\tilde{B}_t\, d\tilde{B}_t$, i.e. $\alpha = 2$ and $\beta = (2B_t, 2\tilde{B}_t)$ against $d(B_t, \tilde{B}_t)$.

(d) $d(t, B_t) = \binom{1}{0} dt + \binom{0}{1} dB_t$.

(e) With $f(t,x) = e^{t/2}\cos x$: $f_t = \frac{1}{2}f$, $f_x = -e^{t/2}\sin x$, $f_{xx} = -f$. Hence

$$d\big(e^{t/2}\cos B_t\big) = \big(\tfrac{1}{2}f - \tfrac{1}{2}f\big)dt - e^{t/2}\sin B_t\, dB_t = -e^{t/2}\sin B_t\, dB_t.$$

So $e^{t/2}\cos B_t = 1 - \int_0^t e^{s/2}\sin B_s\, dB_s$. The integrand is bounded by $e^{T/2}$ on $[0,T]$, so it lies in L^2 and the Itô integral is a martingale. Therefore $e^{t/2}\cos B_t$ is a martingale.

4.

(a) With $f = e^x$, $f' = f'' = f$: $dX_t = e^{B_t}dB_t + \frac{1}{2}e^{B_t}dt = \frac{1}{2}X_t dt + X_t dB_t$.

(b) With $f = \sin x$: $dX_t = \cos B_t\, dB_t - \frac{1}{2}\sin B_t\, dt$. While $\cos B_t > 0$, i.e. for $t < \inf\{s : |B_s| = \pi/2\}$, we have $\cos B_t = \sqrt{1 - \sin^2 B_t} = \sqrt{1 - X_t^2}$. Hence $dX_t = -\frac{1}{2}X_t dt + \sqrt{1 - X_t^2}\, dB_t$.

5. For $k \geq 2$, Itô's formula gives

$$B_t^k = \int_0^t k B_s^{k-1} dB_s + \tfrac{1}{2}k(k-1)\int_0^t B_s^{k-2} ds.$$

The integrand satisfies $\mathbb{E}\int_0^t B_s^{2k-2} ds < \infty$, so the stochastic integral has mean 0. By Fubini,

$$\beta_k(t) = \tfrac{1}{2}k(k-1)\int_0^t \beta_{k-2}(s)\, ds.$$

The formula also holds for $k = 1$ ($\beta_1 = 0$) but fails for $k = 0$, where $\beta_0 = 1$. With $\beta_0 = 1$: $\beta_2(t) = t$, then $\mathbb{E}B_t^4 = \beta_4(t) = 6\int_0^t s\, ds = 3t^2$ and $\mathbb{E}B_t^6 = \beta_6(t) = 15\int_0^t 3s^2\, ds = 15t^3$.

6.

Remark. g_ε is not C^2 on $\mathbb{R}$. It is C^1, with $g_\varepsilon'(x) = \operatorname{sgn} x$ for $|x| \geq \varepsilon$ and x/ε for $|x| < \varepsilon$. Its second derivative $g_\varepsilon'' = \varepsilon^{-1}\mathbf{1}_{(-\varepsilon,\varepsilon)}$ exists except at $\pm\varepsilon$, so g_ε is C^2 on $\mathbb{R} \setminus \{\pm\varepsilon\}$. Itô's formula still applies, as shown in (a).

(a) Let ϕ_n be continuous, $0 \leq \phi_n \leq 1/\varepsilon$, with $\phi_n = g_\varepsilon''$ outside $(\pm\varepsilon - \frac{1}{n}, \pm\varepsilon + \frac{1}{n})$. Let $h_n(x) = g_\varepsilon(0) + \int_0^x \int_0^y \phi_n$, which is C^2. Since $g_\varepsilon'(0) = 0$, $|h_n' - g_\varepsilon'| \leq \int |\phi_n - g_\varepsilon''| \leq 4/(n\varepsilon)$ and $|h_n - g_\varepsilon|(x) \leq 4|x|/(n\varepsilon)$. Itô's formula for h_n reads

$$h_n(B_t) = h_n(B_0) + \int_0^t h_n'(B_s)\, dB_s + \tfrac{1}{2}\int_0^t \phi_n(B_s)\, ds.$$

As $n \to \infty$: $h_n(B_t) \to g_\varepsilon(B_t)$, and $\int h_n'(B)dB \to \int g_\varepsilon'(B)dB$ in L^2 by the isometry and uniform convergence. Also $\phi_n(B_s) \to g_\varepsilon''(B_s)$ for all s with $B_s \neq \pm\varepsilon$, which is a.e. s a.s. because $\mathbb{E}\,\mathrm{Leb}\{s \leq t : B_s = \pm\varepsilon\} = \int_0^t \mathbb{P}(B_s = \pm\varepsilon)ds = 0$. Pass to a subsequence along which the stochastic integrals converge a.s.; bounded convergence for the ds-integrals then gives

$$g_\varepsilon(B_t) = g_\varepsilon(B_0) + \int_0^t g_\varepsilon'(B_s)\, dB_s + \frac{1}{2\varepsilon}\mathrm{Leb}\{s \in [0,t] : B_s \in (-\varepsilon, \varepsilon)\}.$$

(b) By the isometry and dominated convergence ($\mathbb{P}(B_s = 0) = 0$ for $s > 0$),

$$\mathbb{E}\Big(\int_0^t \tfrac{B_s}{\varepsilon}\mathbf{1}_{\{|B_s|<\varepsilon\}}dB_s\Big)^2 = \mathbb{E}\int_0^t \tfrac{B_s^2}{\varepsilon^2}\mathbf{1}_{\{|B_s|<\varepsilon\}}ds \le \int_0^t \mathbb{P}(|B_s|<\varepsilon)\,ds \xrightarrow[\varepsilon\to 0]{} 0.$$

(c) Split $\int g'_\varepsilon(B)dB = \int \operatorname{sgn}(B)\mathbf{1}_{\{|B|\ge\varepsilon\}}dB + \int \frac{B}{\varepsilon}\mathbf{1}_{\{|B|<\varepsilon\}}dB$. The second term tends to 0 in L^2 by (b). The first tends to $\int_0^t \operatorname{sgn}(B_s)dB_s$ in L^2, since $\mathbb{E}\int_0^t \mathbf{1}_{\{|B_s|<\varepsilon\}}ds \to 0$. Moreover $0 \le g_\varepsilon(x) - |x| = (\varepsilon - |x|)^2/(2\varepsilon)\,\mathbf{1}_{\{|x|<\varepsilon\}} \le \varepsilon/2$. Hence by (a)

$$\frac{1}{2\varepsilon}\mathrm{Leb}\{s \le t : B_s \in (-\varepsilon,\varepsilon)\} = g_\varepsilon(B_t) - g_\varepsilon(B_0) - \int_0^t g'_\varepsilon(B_s)dB_s$$

converges in L^2 to $|B_t| - |B_0| - \int_0^t \operatorname{sgn}(B_s)dB_s$. Thus L_t exists as an L^2-limit and $|B_t| = |B_0| + \int_0^t \operatorname{sgn}(B_s)dB_s + L_t$.

7.

Remark. The definition should read $T_x = \inf\{t > 0 : X_t^x = 0\}$ with $x > 0$, not $\inf\{t : X_t^x > 0\}$. In (b) the claim is $(Y_t) \overset{d}{=} (X_t^1)$. All X^x, $x > 0$, are driven by the same B.

Facts used: X^x is continuous and positive on $[0, T_x)$, and $X_t^x \to 0$ as $t \uparrow T_x$ when $T_x < \infty$; also $B_t/t \to 0$ a.s. From the equation we get two more facts.

- On $\{T_x = \infty\}$, X^x is unbounded. Indeed, if $X_t^x < R$ for all $t \ge 0$ then $X_t^x \ge x + B_t + at/R \to \infty$, a contradiction.
- On $\{T_x < \infty\}$, X^x reaches every $r \in (0, x)$.

Hence for $0 < r < x < R$, X^x does not stay in (r, R) forever.

(a) For $t < S = T_x \wedge T_y$, $D_t = X_t^y - X_t^x = y - x + a\int_0^t \big(\frac{1}{X^y} - \frac{1}{X^x}\big)ds$ is C^1 (the integrand is continuous) with $D' = a\frac{X^x - X^y}{X^x X^y} = -aD/(X^x X^y)$. Hence

$$D_t = (y-x)\exp\Big(-a\int_0^t \frac{ds}{X_s^x X_s^y}\Big) > 0 \qquad (t < S).$$

If $T_y < T_x$, then $0 < X_t^x < X_t^y \to 0$ as $t \uparrow T_y$, so $X_{T_y}^x = 0$ by continuity, contradicting $T_y < T_x$. So $T_x \le T_y$ and $X_t^x < X_t^y$ for $t < T_x$. With $y = x$, the same computation gives $D_t = 0 \cdot e^{(\cdots)} = 0$: for a given path of B the equation has at most one positive continuous solution (pathwise uniqueness), and two solutions have the same lifetime.

(b) Let $W_t = x^{-1}B_{x^2 t}$, a BM by scaling. Substituting $s = x^2 u$, $\int_0^{x^2 t} ds/X_s^x = x^2\int_0^t du/X_{x^2 u}^x = x\int_0^t du/Y_u$. Dividing the equation for $X_{x^2 t}^x$ by x gives

$$Y_t = 1 + W_t + a\int_0^t \frac{du}{Y_u}, \qquad t < T_x/x^2.$$

So Y is a positive continuous solution of the equation with initial point 1 driven by W, with lifetime T_x/x^2. By pathwise uniqueness the solution with initial point 1 is a fixed functional of the driving path, $X^1 = \Phi(B)$. Hence $Y = \Phi(W)$ and $Y \overset{d}{=} X^1$ (lifetimes included), since $W \overset{d}{=} B$. This uses that Φ is measurable on path space. That is standard: $Z = X^1 - 1 - B$ solves the pathwise ODE $Z' = a/(1 + w + Z)$, whose solution depends continuously on the path w by Gronwall's inequality. We take it for granted, as the problem does in asserting that the solution is well defined.

(c) Let $s(x) = x^{1-2a}$ for $a \ne \frac{1}{2}$ and $s(x) = \log x$ for $a = \frac{1}{2}$. Then $s'(x) = c\,x^{-2a}$ with $c \ne 0$, and $\frac{1}{2}s'' + \frac{a}{x}s' = 0$. For $0 < r < x < R$ let τ be the exit time of X^x from (r, R); it is a.s. finite by the facts above. On $[0, \tau]$, X^x is an Itô process $dX = \frac{a}{X}dt + dB$ with values in $[r, R]$, and s is smooth there. So

Itô's formula gives $s(X_{t\wedge\tau}) = s(x) + \int_0^{t\wedge\tau} s'(X)dB$, with integrand bounded by $\max_{[r,R]}|s'|$, hence a martingale. Bounded convergence ($t \to \infty$) yields $\mathbb{E}s(X_\tau) = s(x)$, i.e.

$$p_{r,R}(x) := \mathbb{P}(X^x \text{ hits } r \text{ before } R) = \frac{s(R) - s(x)}{s(R) - s(r)}.$$

By (a), for $x < y$ we have $T_x \le T_y$ and $X^x < X^y$ on $[0, T_x)$. Properties ii and iv pass from y to every $x < y$, and properties i and iii pass from x to every $y > x$. So it suffices to prove ii and iv for $x \in \mathbb{N}$, and i and iii for $x = 1/n$, $n \in \mathbb{N}$. These are countably many values, so the almost sure statements hold simultaneously for all $x > 0$.

i. ($a \ge \frac{1}{2}$.) If $T_x < \infty$, the path is continuous on $[0, T_x]$, so $\sup_{[0,T_x]} X^x < R$ for some $R \in \mathbb{N}$. Then X^x hits every $r \in (0, x)$ before R. Now $p_{r,R}(x) \to 0$ as $r \to 0$: for $a = \frac{1}{2}$, $s(r) = \log r \to -\infty$; for $a > \frac{1}{2}$, $s(r) \to +\infty$ while $s(R) - s(x)$ is finite. Hence $\mathbb{P}(T_x < \infty) \le \sum_{R\in\mathbb{N},\, R>x} \lim_{r\to 0} p_{r,R}(x) = 0$.

ii. ($a = \frac{1}{2}$.) By i, $T_x = \infty$. For $0 < r < x < R$,

$$\mathbb{P}(X^x \text{ hits } r) \ge p_{r,R}(x) = \frac{\log R - \log x}{\log R - \log r} \xrightarrow[R\to\infty]{} 1.$$

Intersecting over $r = 1/n$ gives $\inf_{t>0} X_t^x = 0$ a.s.

iii. ($a > \frac{1}{2}$.) By i, $T_x = \infty$. Let $M_t = (X_t^x)^{1-2a} > 0$ and $\sigma_n = \inf\{t : X_t^x \notin (\frac{1}{n}, n)\}$. Then $\sigma_n \uparrow \infty$, since the path is continuous and positive on $[0, \infty)$, and $M_{t\wedge\sigma_n} = M_0 + \int_0^{t\wedge\sigma_n} s'(X)dB$ is a bounded martingale. Conditional Fatou gives $\mathbb{E}[M_t|\mathcal{F}_s] \le \liminf_n \mathbb{E}[M_{t\wedge\sigma_n}|\mathcal{F}_s] = \liminf_n M_{s\wedge\sigma_n} = M_s$ (in particular $\mathbb{E}M_t \le M_0$). So M is a non-negative continuous supermartingale. By Exercise 2, Martingales 5(a) applied to the submartingale $-M$ (with $\sup_t \mathbb{E}(-M_t)^+ = 0$), $M_t \to M_\infty \in [0, \infty)$ a.s. Hence $X_t^x \to X_\infty \in (0, \infty]$. If $X_\infty < \infty$, the continuous path would be bounded on $[0, \infty)$, contradicting the first fact. So $X_t^x \to \infty$ a.s.

iv. ($a < \frac{1}{2}$.) Here $s(0^+) = 0$ and $s(\infty) = \infty$. Fix $R > x$ and let τ_r ($r < x$) and τ_R be the hitting times of r and R ($\inf\emptyset = \infty$). The events $E_r = \{\tau_r < \tau_R\}$ decrease as $r \downarrow 0$. We claim $\bigcap_r E_r = \{T_x < \tau_R\}$ up to a null set.

The inclusion $\supseteq$ holds by continuity. Conversely, on $\bigcap_r E_r$ the times τ_r increase to some $\sigma \le T_x$, and $\sigma < \infty$ a.s.: if $T_x = \infty$ then $\tau_R < \infty$ by the first fact, and $\sigma \le \tau_R$. If $\sigma < T_x$, continuity at σ would give $X_\sigma^x = \lim_r X_{\tau_r}^x = 0$, contradicting positivity on $[0, T_x)$. So $\sigma = T_x$. Since $\tau_r < \tau_R$ for all r, X^x does not reach R before T_x, i.e. $T_x < \tau_R$. Hence

$$\mathbb{P}(T_x < \tau_R) = \lim_{r\to 0} p_{r,R}(x) = \lim_{r\to 0} \frac{R^{1-2a} - x^{1-2a}}{R^{1-2a} - r^{1-2a}} = 1 - (x/R)^{1-2a} \xrightarrow[R\to\infty]{} 1,$$

so $T_x < \infty$ a.s.

Setting for v and vi. Fix $y > x > 0$ and write $X = X^x$, $D = X^y - X^x$. By (a) and iv, $T_x \le T_y$, $T_x < \infty$ a.s., and D is C^1, positive and non-increasing on $[0, T_x)$. Let $D_{T_x} := \lim_{t\uparrow T_x} D_t \ge 0$. Since $X_t^y = X_t + D_t \to D_{T_x}$ as $t \uparrow T_x$, we have $T_x = T_y$ iff $D_{T_x} = 0$. Moreover, by (a),

$$\log D_{T_x} = \log(y - x) - a\int_0^{T_x} \frac{dt}{X_t X_t^y}. \tag{$\dagger$}$$

On $[0, T_x)$ let $V = X/D > 0$, so that $X = DV$ and $X^y = D(V+1)$. Since $1/D$ is C^1 with $d(1/D) = \frac{a}{XX^yD}dt$ (no quadratic covariation with X), the product rule gives

$$dV = \frac{1}{D}\Big(\frac{a}{X}dt + dB\Big) + X\,d\Big(\frac{1}{D}\Big) = \frac{1}{D^2}\Big(\frac{a}{V} + \frac{a}{V+1}\Big)dt + \frac{1}{D}\,dB, \qquad d\langle V\rangle_t = \frac{dt}{D_t^2}.$$

Hence for $\varphi \in C^2(0, \infty)$, Itô's formula gives $d\varphi(V) = D^{-2}\big(\frac{1}{2}\varphi'' + (\frac{a}{V} + \frac{a}{V+1})\varphi'\big)(V)\,dt + \varphi'(V)D^{-1}dB$. Let

$$S(v) = \int_0^v \big(u(1+u)\big)^{-2a}du, \qquad v \ge 0,$$

which is finite since $2a < 1$, increasing, and has decreasing derivative $S'(v) = (v(1+v))^{-2a}$. Since $S''/S' = -2a(\frac{1}{v} + \frac{1}{1+v})$, we get $\frac{1}{2}S'' + (\frac{a}{v} + \frac{a}{v+1})S' = 0$, so

$$dS(V) = S'(V)D^{-1}dB.$$

For $n \in \mathbb{N}$ let $\rho_n = \inf\{t < T_x : D_t \le \frac{1}{n} \text{ or } V_t \le \frac{1}{n}\} \wedge n$.

$\rho_n < T_x$. If $D_{T_x} < \frac{1}{n}$, then D reaches $\frac{1}{n}$ strictly before T_x. Otherwise $D \ge \frac{1}{n}$ on $[0, T_x)$, so $V \le nX \to 0$ and V reaches $\frac{1}{n}$ strictly before T_x. Thus $V_{\rho_n \wedge \kappa}$ is well defined for every stopping time κ.

Optional stopping. On $[0, \rho_n]$, $|S'(V)D^{-1}| \le S'(\frac{1}{n})\, n$ and $\rho_n \le n$. Hence $\mathbb{E}S(V_{\rho_n \wedge \kappa}) = S(V_0)$ for every stopping time κ.

$\rho_n \uparrow T_x$. (ρ_n) is non-decreasing. If its limit ρ_∞ were $< T_x$, then D and V would be continuous and positive on the compact interval $[0, \rho_\infty]$, hence $\ge c > 0$ there. For $n > \max(1/c, \rho_\infty)$ this gives $\rho_n > \rho_\infty$, a contradiction.

v. ($\frac{1}{4} < a < \frac{1}{2}$.) Now $S(\infty) < \infty$, since $(u(1+u))^{-2a} \le u^{-4a}$ and $4a > 1$. Suppose, for contradiction, that $\mathbb{P}(T_x < T_y) = 1$. Then a.s. $D_{T_x} > 0$ and $X_t \to 0$, so $V_t \to 0$ as $t \uparrow T_x$. For $n > \max(1/D_{T_x}, T_x, 1/V_0)$ we have $D > \frac{1}{n}$ on $[0, T_x)$ and $T_x < n$, so ρ_n is the hitting time of $\frac{1}{n}$ by V, and $S(V_{\rho_n}) = S(\frac{1}{n}) \to S(0) = 0$. Since $0 \le S \le S(\infty)$, bounded convergence gives $S(V_0) = \lim_n \mathbb{E}S(V_{\rho_n}) = 0$. This contradicts $V_0 = x/(y-x) > 0$. Hence $\mathbb{P}(T_x = T_y) > 0$.

vi. ($a \le \frac{1}{4}$.) Now $S(\infty) = \infty$, since for $u \ge 1$ we have $u(1+u) \le 2u^2$, hence $(u(1+u))^{-2a} \ge 2^{-2a}u^{-4a} \ge 2^{-2a}u^{-1}$ (as $4a \le 1$), which is not integrable at ∞.

Step 1: $\sup_{t<T_x} V_t < \infty$ *a.s.* For $\lambda > S(V_0)$ let $\kappa_\lambda = \inf\{t < T_x : S(V_t) \ge \lambda\}$ (with $\kappa_\lambda = T_x$ if this set is empty). On $\{\kappa_\lambda \le \rho_n\}$ continuity gives $S(V_{\kappa_\lambda}) = \lambda$. As $S \ge 0$, $\lambda\mathbb{P}(\kappa_\lambda \le \rho_n) \le \mathbb{E}S(V_{\rho_n \wedge \kappa_\lambda}) = S(V_0)$. Since $\rho_n \uparrow T_x$, every ω with $\kappa_\lambda < T_x$ satisfies $\kappa_\lambda \le \rho_n$ for large n. Hence $\mathbb{P}(\kappa_\lambda < T_x) \le S(V_0)/\lambda \to 0$ as $\lambda \to \infty$. As S is increasing with $S(\infty) = \infty$, this gives $\sup_{t<T_x} V_t < \infty$ a.s.

Step 2: $D_{T_x} > 0$ *on* $\{\sup_{t<T_x} V_t < K\}$*, for each* $K \in \mathbb{N}$ *with* $K > V_0$. Let $\rho'_n = \inf\{t < T_x : D_t \le \frac{1}{n} \text{ or } V_t \ge K\}$ (with $\inf\emptyset = T_x$). For $m \ge n$ put $\theta = \rho'_n \wedge \rho_m$. Then $\theta < T_x$, $\theta \le m$, $D \ge \frac{1}{m}$ on $[0, \theta]$, and $0 < V \le K$ on $[0, \theta]$. Integrating the equation for V up to θ, the dB-integral has bounded integrand $1/D \le m$ and bounded time horizon, so it has mean 0. Dropping the non-negative term $\frac{a}{V+1}$,

$$\mathbb{E}\int_0^\theta \frac{a}{V_t D_t^2}\, dt \le \mathbb{E}V_\theta - V_0 \le K.$$

As $m \to \infty$, $\theta \uparrow \rho'_n$ (because $\rho_m \uparrow T_x \ge \rho'_n$), so by monotone convergence $\mathbb{E}\int_0^{\rho'_n} \frac{a\,dt}{V_t D_t^2} \le K$. On $\{\sup_{t<T_x} V < K\}$, ρ'_n is the hitting time of $\frac{1}{n}$ by D, which increases to T_x as $n \to \infty$ (D is continuous and positive on $[0, T_x)$). Monotone convergence again gives $\mathbb{E}\big[\mathbf{1}_{\{\sup V<K\}} \int_0^{T_x} \frac{dt}{V_t D_t^2}\big] \le K/a < \infty$. Since $\frac{1}{XX^y} = \frac{1}{D^2V(V+1)} \le \frac{1}{D^2V}$, (†) yields $D_{T_x} > 0$ a.s. on $\{\sup_{t<T_x} V < K\}$.

Conclusion. By Step 1, $\bigcup_{K\in\mathbb{N},\, K>V_0}\{\sup_{t<T_x} V < K\}$ has probability 1. So $D_{T_x} > 0$ a.s., i.e. $T_x < T_y$ a.s.